

Samantha Creech

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Education

- **PhD candidate**, University of Utah, *August 2021-May 2026*
 - **MS of Physics**, University of Utah, *August 2023*
 - **BS in Physics, summa cum laude** University of North Carolina at Asheville, *May 2020*
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Honors and Awards

- **NASA FINESST fellow:** earned for my graduate research 2024
 - **Swigart fellow:** earned for summer research in the first year of my graduate career 2022
 - **EDGES fellow:** earned for my commitment to bettering the culture of STEM for underrepresented groups 2021
 - **Manly E Wright Award:** UNC Asheville's valedictorian award 2020
 - **Distinction as a University Scholar:** Earned for my excellent academic record 2020
 - **Astronomy Club of Asheville-Carolyn Keefe Scholarship:** earned for my academic, research, and outreach-related accomplishments in astronomy 2019
 - **North Carolina Space Grant:** earned for my research in astronomy 2018
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Research Experience

Research Assistant *January 2022-Present*

University of Utah, Salt Lake City, UT

- Analyzed X-ray spectra from a survey of faint Active Galactic Nuclei and compared to population synthesis models, resulting in an accepted first-author publication
- Used IDL, Xspec, and Python to analyze *NuSTAR* X-ray data of the Perseus Cluster, presented in a first-authored paper
- Calculated upper limit on Inverse Compton scattering emission in the Perseus Cluster, resulting in estimate of magnetic field strength
- Characterized the systematic uncertainty for *nucrossarf*, a software suite that accounts for scattered light in spectral analysis, developing a novel technique for addressing similar uncertainties that will be useful for future work

Astrophysics REU Participant *May-August 2019*

University of Wisconsin-Madison, WI

- Performed laboratory tests of optical fibers for use in the next generation of neutrino telescopes, which helped inform designs for future detectors
- Built and ran particle simulations using C++ and Geant4, providing estimates of optical fiber efficiency
- Analyzed and compared laboratory and simulation data using Python, C++, and ROOT
- Presented results at the UNCA Undergraduate Research Symposium (November 2019), and the American Astronomical Society meeting in Honolulu (January 2020)

Research Assistant August 2017-May 2021

University of North Carolina at Asheville, NC

- Used Python and SAOImage DS9 to analyze Hubble Space Telescope infrared images
- Analyzed galaxy morphology using Galfit and SAOImage DS9, which became a key result in a second-authored paper
- Presented results at the UNCA Undergraduate Research Symposium (November 2018), the American Astronomical Society meeting in Seattle (January 2019), and the UNCA Foundation Board meeting (December 2019)

Professional Publications

Creech S, Civano S, Wik D. et al. The N_H Distribution of Hard X-ray Selected AGN in the NEP Field .
Accepted by the Astrophysical Journal. arXiv:2510.26554

- **Contribution:** Performed comprehensive analysis and authored the entirety of the paper, with feedback, suggestions, and supplemental analysis from co-authors

Silver R, Civano F, Zhao X, **Creech S**, et al. PEARLS: PEARLS: NuSTAR and XMM-Newton Extragalactic Survey of the JWST North Ecliptic Pole Time-domain Field III. *Accepted by the Astrophysical Journal*. arXiv:2510.24858

- **Contribution:** Collaborated and provided feedback through engagement in weekly meetings

Civano F, Zhao X, Boorman P, Marches S, Ananna T, **Creech S**, et al. The High Energy X-ray Probe (HEX-P): Bringing the Cosmic X-ray Background into focus. *Accepted by Frontiers in Astronomy and Space Science*. arXiv:2311.04832

- **Contribution:** Performed analysis and contributed text for section 6.1

Boorman P, Torres-Albà N, Annular A, Marchesi S, Pfeifle R, Stern D, Civano F, Baloković M, Buchner J, Ricci C, Alexander D. M, Brandt W. N, Brightman M, Chen C. T, **Creech S**, et al. The High Energy X-ray Probe (HEX-P): The Circum-nuclear Environment of Growing Supermassive Black Holes. *Frontiers in Astronomy and Space Sciences*, vol. 11, id. 1335459. arXiv:2311.04832

- **Contribution:** Collaborated through engagement in bi-weekly meetings

Zhao X, Civano F, Willmer C, Bonoli S, Chen C, **Creech S**, et al. PEARLS: NuSTAR and XMM-Newton Extragalactic Survey of the JWST North Ecliptic Pole Time-Domain Field II. *The Astrophysical Journal*, Volume 965, Issue 2, id.188, 29 pp. arXiv:2401.16616

- **Contribution:** Collaborated through engagement in weekly meetings

Creech S, Wik D, et al. The NuSTAR View of Perseus: the ICM and a Peculiar Hard Excess. *The Astrophysical Journal*, Volume 965, Issue 1, id.33, 16 pp. arXiv:2401.16616

- **Contribution:** Performed comprehensive analysis and authored the entirety of the paper, with feedback and suggestions from co-authors

Lundgren B, **Creech S** et al. The Geometry Of Cold, Metal-Enriched Circumgalactic Gas Around Galaxies at $z \sim 1.2$. *The Astrophysical Journal*, v913, p50, 202. arXiv:2102.10117

- **Contribution:** Created ~half of the figures and performed analysis/contributed text for sections 4.5 and 4.6

Selected Professional Presentations

Talks

Invited talk: Spectral Analysis of Hard X-ray Selected AGN in the North Ecliptic Pole Field: NH distribution and Variability. Talk to be presented at: Accretion History of AGN workshop; 2025 December 15; Miami, FL.

Spectral Analysis of Hard X-ray Selected AGN in the North Ecliptic Pole Field: NH distribution and Variability. Talk presented at: HEAD22 meeting; 2025 October 14; St Louis, MO.

Spectral Analysis of Hard X-ray Selected AGN in the North Ecliptic Pole Field: NH distribution and Variability. Talk presented at: HEAD22 meeting; 2025 October 14; St Louis, MO.

Measuring Dust around Supermassive Black Holes in a Deep X-ray Survey. Talk presented at: Caltech Astronomy Group Tea Talk Seminar; 2025 Sep 3rd Pasadena CA and presented again at Clemson University seminar; 2025 October 4th

X-ray Spectral Properties of AGN in the NuSTAR-JWST North Ecliptic Pole Survey: First Results and the Compton Thick Fraction. Talk presented at: 245th American Astronomical Society conference; 2025 Jan 12-16; National Harbor, MD.

Evidence of a Hard Excess in NuSTAR Spectra of the Perseus Cluster's Diffuse Intracluster Medium. Talk presented at: HEAD20 meeting; 2023 March 29; Waikōloa, HI.

Posters

Spectral Analysis of Hard X-ray Selected AGN in the North Ecliptic Pole (NEP) Field. Poster presented at: Vasto Accretion Meeting; 2025 June 30-July 4; Vasto, Italy.

The 5 Year Monitoring Campaign of the JWST Time-Domain Field in the NEP with NuSTAR and XMM-Newton. Poster presented at: 243rd American Astronomical Society conference; 2024 Jan 7-11; New Orleans, LA.

Conferences Attended

- High Energy Astrophysics Division Meeting (HEAD)- St Louis, Missouri, October 2025
- Vasto Accretion Meeting- Vasto, Italy, July 2025
- American Astronomical Society (AAS)- National Harbor, Maryland, January 2025
- American Astronomical Society (AAS)- New Orleans, Louisiana, January 2024
- High Energy Astrophysics Division Meeting (HEAD)- Waikōloa, Hawai'i, March 2023
- American Astronomical Society (AAS)- Virtual, June 2020
- Conference for Undergraduate Women in Physics (CUWiP)- College Park, Maryland, January 2020
- American Astronomical Society (AAS)- Honolulu, Hawai'i, January 2020
- American Astronomical Society (AAS)- Seattle, Washington, January 2019

Workshops Attended

- Center for Teaching Excellence 2025 Graduate Teaching Institute- University of Utah, Salt lake City, UT, May 2025
- Data Visualization Workshop- AAS meeting, National Harbor, MD, January 2025
- Science Communication Workshop Series- Natural History Museum of Utah, Salt Lake City, UT, March-May 2025

Teaching and Outreach Experience

Research Mentor *May 2025-Present*

University of Utah, Salt Lake City, UT

- Guided student into a brand new field and taught relevant technical skills, resulting in her successfully analyzing the X-ray spectra from an entire survey of AGN within a summer
- Advised student on career plans, and will be helping her apply to graduate schools in the fall
- Set up weekly one-on-one meetings, allowing student's questions to be addressed quickly and enabling her swift progress

Adjunct faculty *May 2025-August 2025*

Salt Lake Community College, Salt Lake City, UT

- Taught introductory astronomy to diverse student body with a 91% pass rate
- Utilized active learning methods to increase engagement and success, resulting in over half the class asking regular questions and participating in discussion
- Fostered a positive learning environment, which created an encouraging culture between students
- Organized an optional planetarium trip for students that was attended by half of the class

Science Communication Fellowship Program participant *January 2024-January 2025*

Natural History Museum of Utah, Salt Lake City, UT

- Completed science communication workshop series, developing broadly applicable communication skills
- Designed *Scientist In the Spotlight* activities, resulting in engagement from visitors of all ages
- Designed *Science Cafe* research talk for general public, which was highly attended and received positive feedback from visitors

Volunteer Gallery Interpreter *April 2023-Present*

Natural History Museum of Utah, Salt Lake City, UT

- Planned interactive activities for museum visitors of all ages, which resulted in all family members showing excitement for learning
- Interpreted paleontology displays in the Past Worlds exhibit, enhancing visitors' experience of the museum

Physics Lab Teaching Assistant *August -December 2021*

University of Utah, Salt Lake City, UT

- Provided weekly lectures to introduce each lab for PHYS 2220: the introductory physics lab for scientists and engineers
- Facilitated student discussion and conceptual understanding during lab activities, resulting in 92% students who filled out evaluations reporting that I had a positive impact on their ability to ask questions
- Graded all student assignments and provided thorough feedback, improving students' scientific communication throughout the semester

Intern Interpretive Dark-Sky Ranger *May -August 2021*

Bryce Canyon National Park, Bryce City, UT

- Designed engaging interpretive astronomy programs such as constellation tours, a full-moon interpretive night hike, and a 40-minute ranger talk, which were highly attended and received positive feedback
- Built an interactive light-pollution model to educate school groups on dark-sky-friendly lighting
- Interacted with visitors and communicated important park information, increasing visitor safety
- Used scientific equipment to monitor light pollution and wrote the park's annual dark sky report, which helped to secure the park's dark-sky status for the following year.

Physics Tutor *August 2019-May 2020*

University of North Carolina at Asheville

- Provided in-person and remote instruction to students in introductory physics classes, improving students' understanding of physics
- Assisted students in the transition to remote learning

In Real Life (IRL) After-School Program Leader *September-November 2019*

Asheville Middle School, Asheville, NC

- Planned and led lab activities for IRL, which is an after-school program for at-risk middle school students, resulting in increased student interest in STEM
- Taught students basic topics from astronomy including spectroscopy, telescope optics, and stellar evolution
- Supervised the students during labs and activities, ensuring that labs were safe and enjoyable to all

Teaching Assistant *January-May 2020*

University of North Carolina at Asheville

- Created and executed lesson plans for the Outdoor Leadership Training Program, increasing engagement within the class
- Planned and led a five-day climbing training for the twelve students in the class, resulting in the next generation of skilled trip leaders

Observatory Docent *August 2017-May 2020*

UNCA Lookout Observatory, Asheville, NC

- Operated computerized telescopes and SkyX software, resulting in safe and successful public stargaze events
- Taught astronomy and physics concepts to the general public, increasing scientific knowledge for community members
- Led outreach activities for summer camps, school groups, and on-campus programs, increasing interest in STEM for young students

Educational Publications and Presentations

"How Many Chickens Are in the Universe and Other April Fools Projects" Talk presented at: Salt Lake City Astronomy on Tap; April 15, 2025

"X-Raying the Centers of Galaxies" Talk presented at: Natural History Museum of Utah Science Cafe; November 13, 2024

"25 Years of NASA's *Chandra* X-ray Observatory" Talk presented at: Salt Lake City Astronomy on Tap; July 11, 2024

"Volcanoes of the Solar System" Talk presented at: Salt Lake City Astronomy on Tap; October 17, 2023 and Capitol Reef Heritage StarFest; September 19, 2025

"How Stars Connect Us" Constellation tour presented weekly at: Bryce Canyon National Park; June-August 2021

"Pushing the Horizon: How astronomers came together to take the first picture of a black hole" Talk presented weekly at: Bryce Canyon National Park; June-August 2021 and presented again at Salt Lake City Astronomy on Tap; March 21 2023

Creech S. "Io's Stealthy Volcanoes." *Physics Today Daily Edition*, August 18 2020.

DOI:10.1063/PT.6.1.20200819a

Service Experience

Graduate Research Seminar (GRS) Coordinator *August 2025-Present*

University of Utah

- Organized, scheduled, and facilitated weekly seminar for graduate students to practice talks and give one another feedback, resulting in high engagement between students and increased success at talks

Graduate Student Advisory Committee (GSAC) Chair *June 2024-May 2025*

University of Utah

- Worked with department leadership to advocate on behalf of graduate student issues, resulting in better conditions for students
- Conducted surveys for graduate students, analyzed responses, and presented results to faculty and department leadership, which increased awareness about students' financial struggles
- Planned, organized, and hosted GSAC open door meetings and graduate student town halls with high attendance compared to previous years, increasing student engagement in student advocacy

Graduate Student Advisory Committee (GSAC) Treasurer, Member, and Chair Elect *June 2022-June 2024*

University of Utah

- Presented monthly budgetary updates to the committee, keeping GSAC's finances on track
- Trained to become chair, resulting in a successful year guiding GSAC

Student Leader for Physics and Astronomy Society for Support and Advocacy for Gender Equity (PASSAGE)

August 2022-May 2023

University of Utah

- Planned, organized, and hosted social events with high attendance
- Facilitated the development of community among gender minorities in the department
- Assisted hosting professional development workshops, which helped members develop skills and knowledge to succeed in their careers

Society of Physics Students Chapter President *August 2019-May 2020*

University of North Carolina at Asheville

- Led officer and club meetings and events with high engagement from students
- Organized physics outreach training for new club members, increasing the outreach participation
- Organized and lead physics demonstrations for visiting middle school classes, sparking youth interest in STEM

Professional Experience

Society of Physics Students Intern: Physics Today Science Writer *June-August 2020*

American Institute of Physics, College Park, MD

- Reviewed scientific journals to find stories for *Physics Today*, a magazine that targets the physics community
- Interviewed lead researchers for stories
- Wrote and published articles in *Physics Today* that successfully communicate technical research

Information Technology intern *June-August 2017*

Pisgah Astronomical Research Institute (PARI), Rosman, NC

- Created three-dimensional models of planetary features using Python and Blender, resulting in 3D-printed models for **accessible science communication**
 - Improved a python-blender script that uses planetary elevation data to make three-dimensional models, adding three planetary bodies (Ceres, Phobos, and Demos) to the script's capabilities
 - Created an interpretive video for 3D Planets educational program, increasing visitor awareness of the program and its accessibility goals
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Other Experience

NOLS Student *September 2020-December 2020*

NOLS Rocky Mountain branch

- Developed advanced leadership skills in a high-stress environment, which revolutionized my leadership and presentation abilities in all settings
- Planned and executed a logistically challenging, week-long independent student group trip, demonstrating my ability to use new skills without supervision

Outdoor Programs: Trip Leader and Office Supervisor *August 2017-May 2020*

UNCA Campus Recreation, Asheville, NC

- Taught skills to participants during climbing, backpacking, caving, and mountain biking trips, resulting in safe and successful experiences
- Developed training programs for Outdoor Programs office staff, which reduced logistical and budgetary issues
- Led incoming students during wilderness orientation trips for three years, which increased first-years' sense of belonging during the transition into college